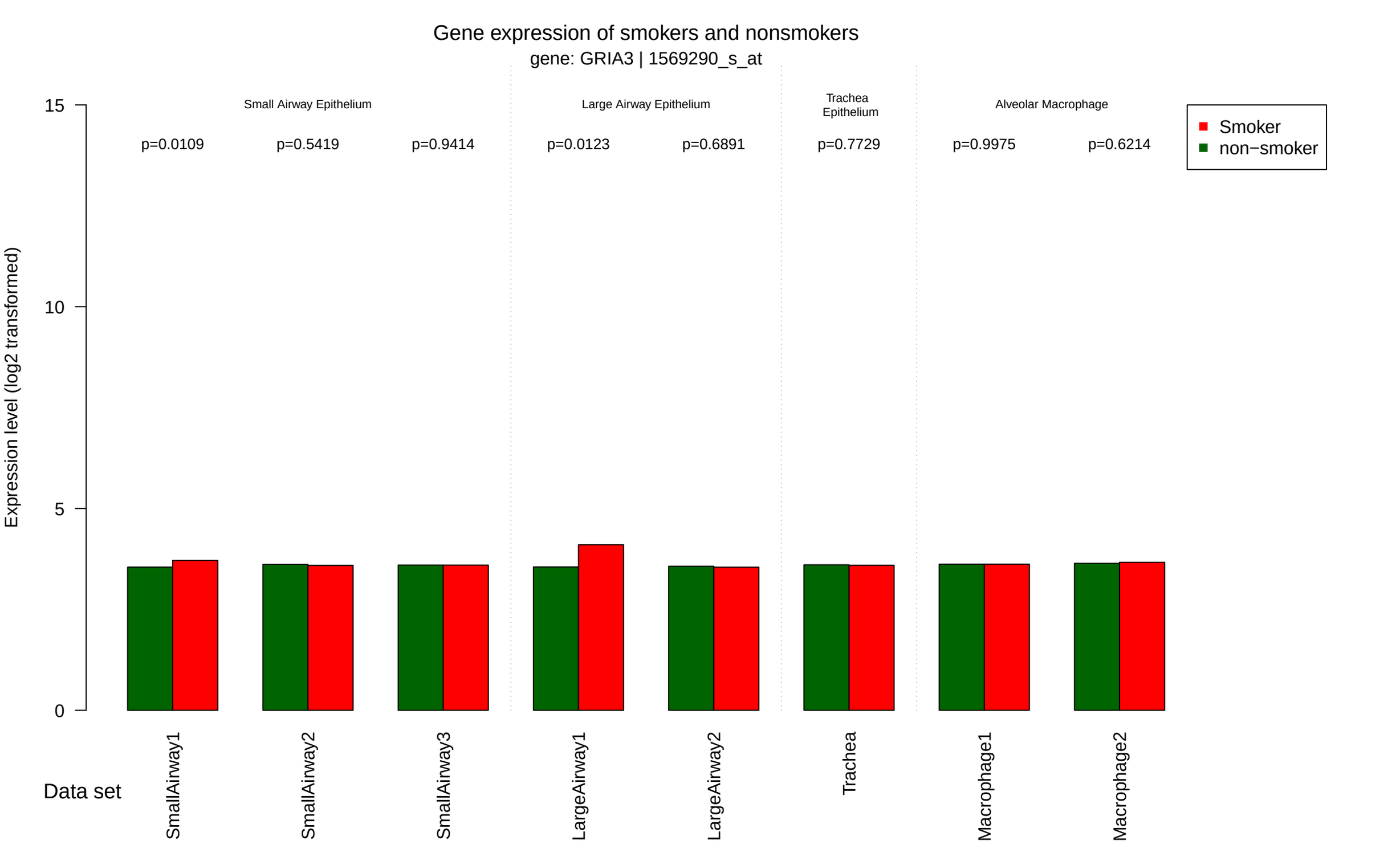
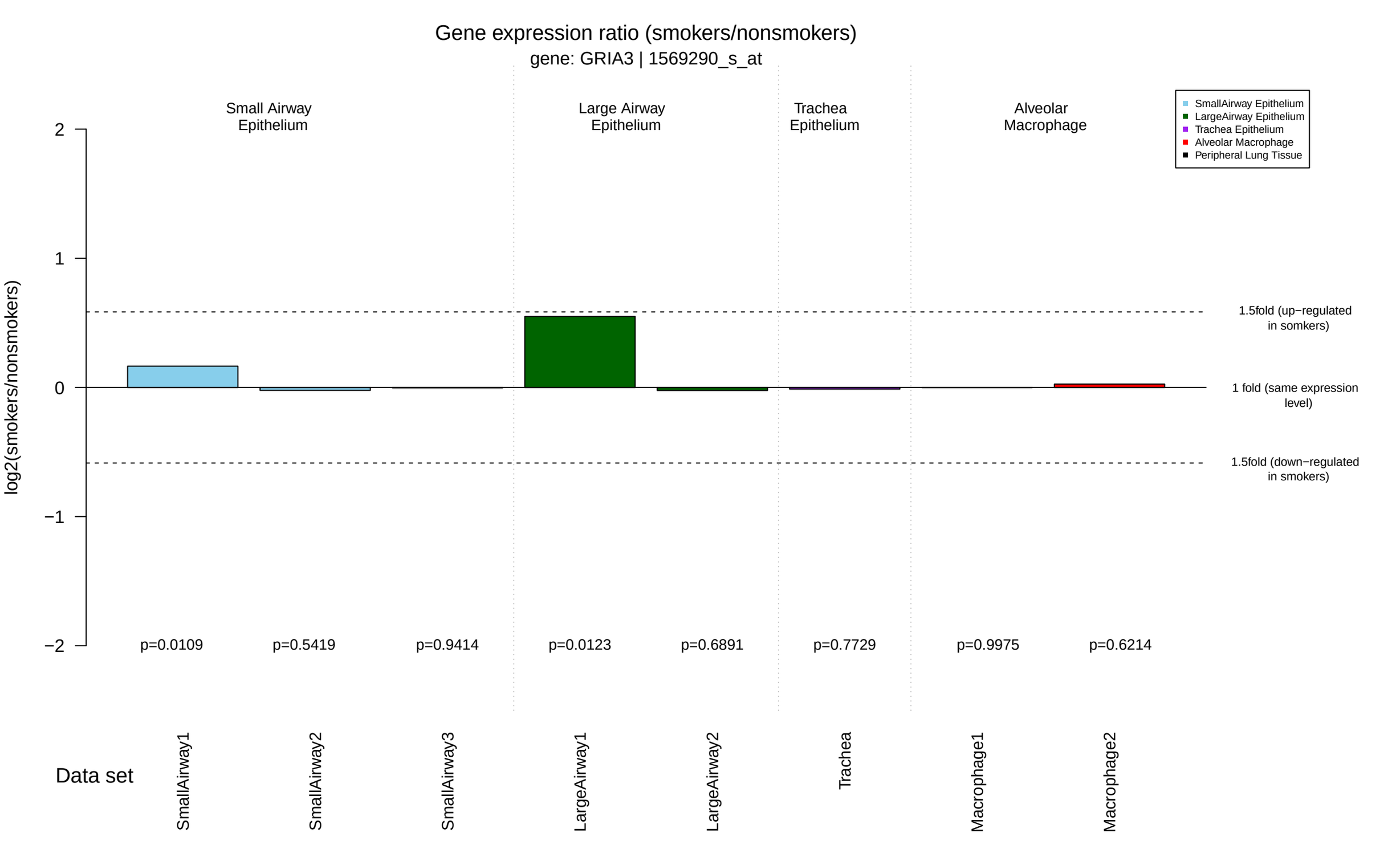
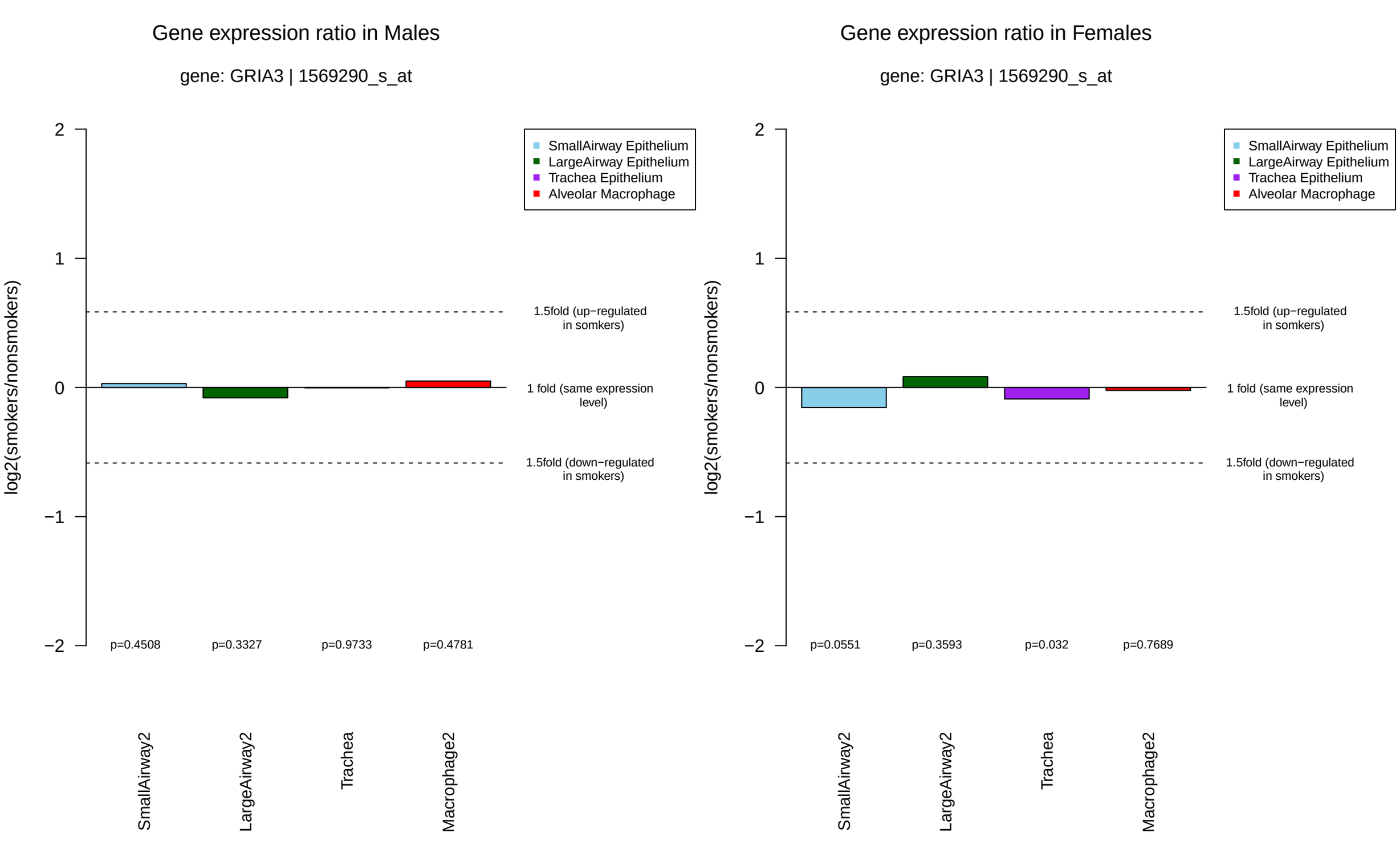
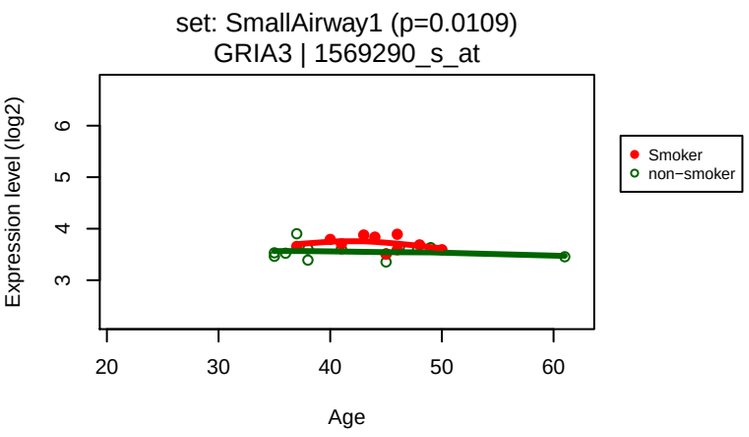
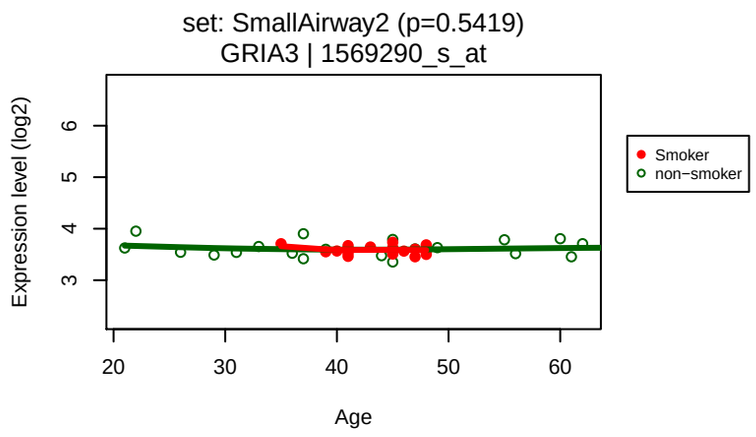
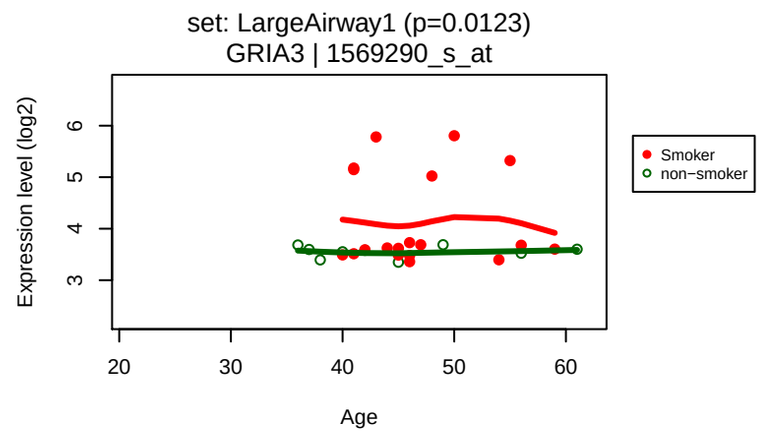

Gene expression of smokers and nonsmokers
gene: GRIA3 | 1569290_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0109
p=0.5419
p=0.9414
p=0.0123
p=0.6891
p=0.7729
p=0.9975
p=0.6214
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

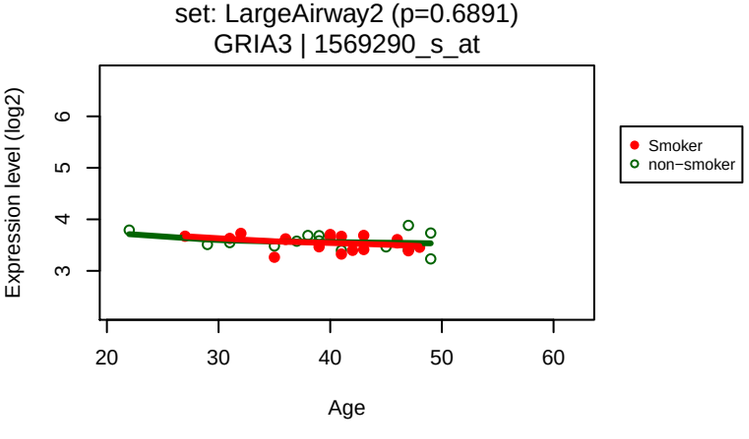

Gene expression ratio (smokers/nonsmokers)
gene: GRIA3 | 1569290_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0109
p=0.5419
p=0.9414
p=0.0123
p=0.6891
p=0.7729
p=0.9975
p=0.6214
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

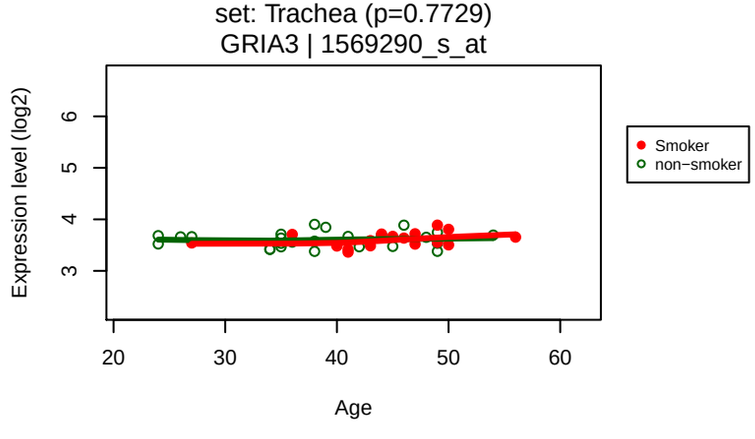

Gene expression ratio in Males
gene: GRIA3 | 1569290_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4508
p=0.3327
p=0.9733
p=0.4781
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRIA3 | 1569290_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0551
p=0.3593
p=0.032
p=0.7689
SmallAirway2
LargeAirway2
Trachea
Macrophage2

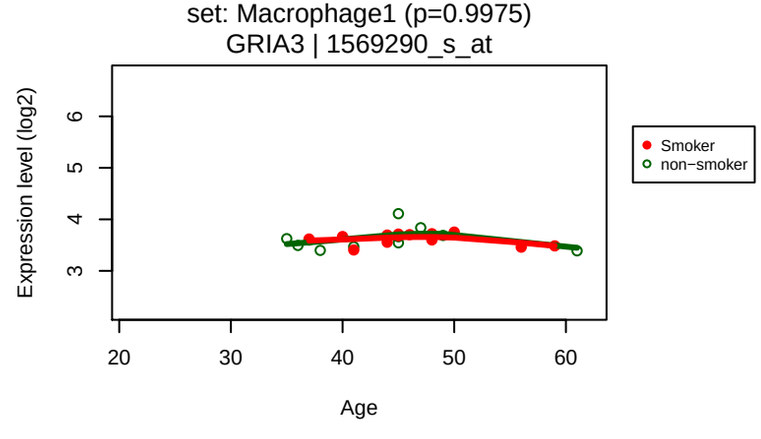

set: SmallAirway1 (p=0.0109)
GRIA3 | 1569290_s_at
Expression level (log2)
Age
Smoker
non−smoker

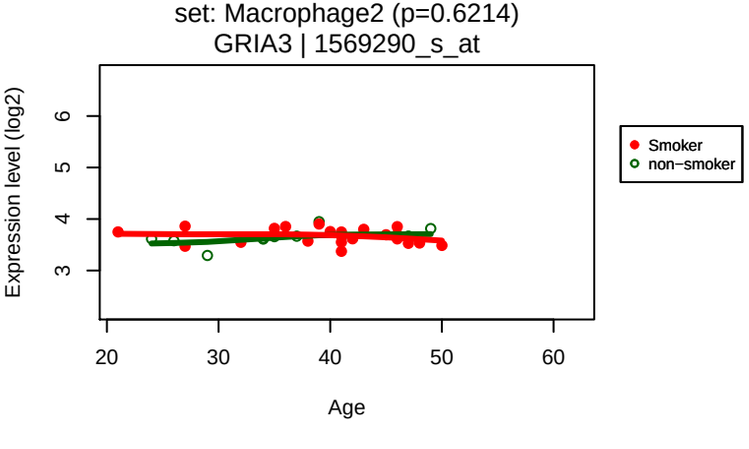

set: SmallAirway2 (p=0.5419)
GRIA3 | 1569290_s_at
Expression level (log2)
Age
Smoker
non−smoker

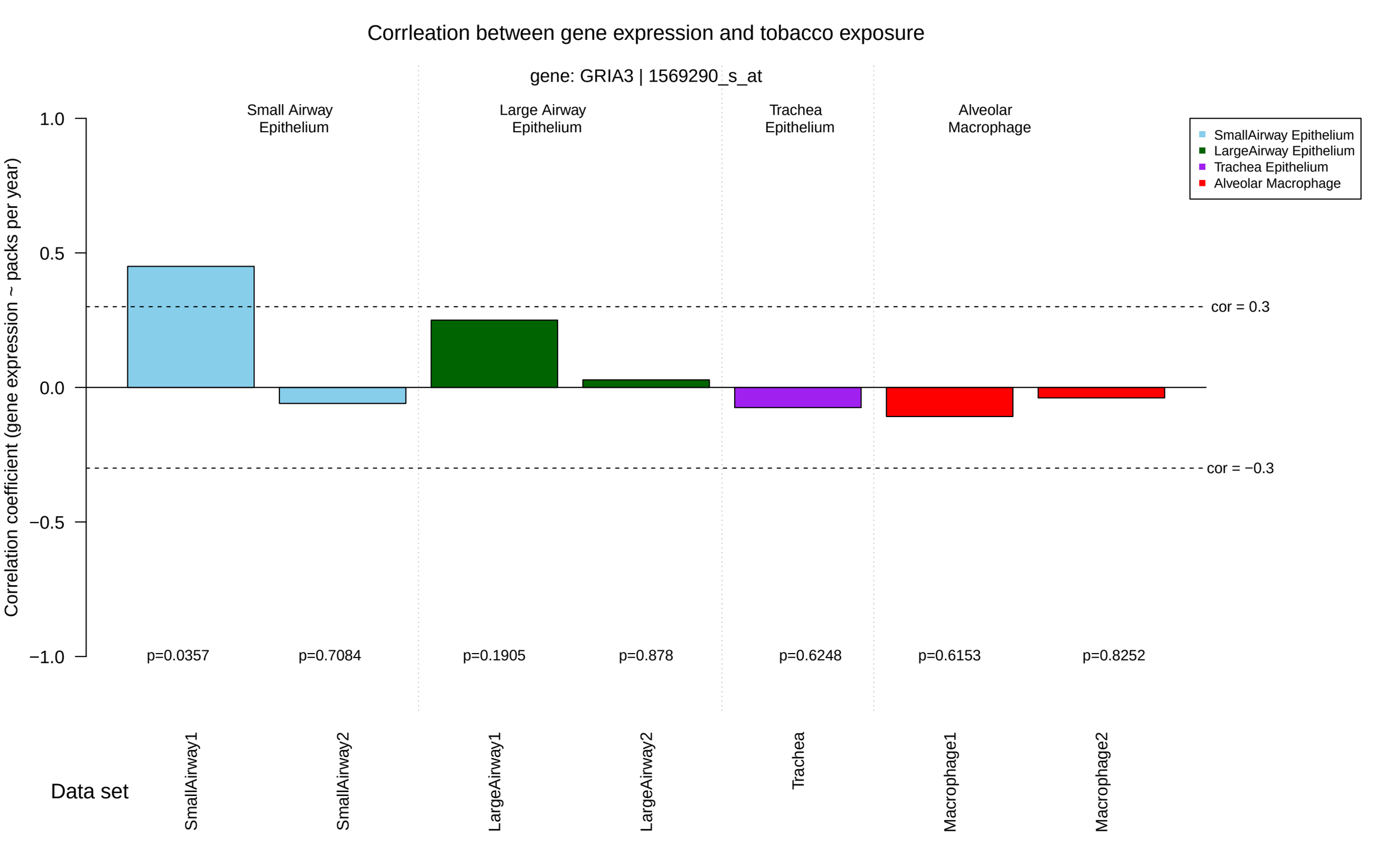

set: LargeAirway1 (p=0.0123)
GRIA3 | 1569290_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6891)
GRIA3 | 1569290_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7729)
GRIA3 | 1569290_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9975)
GRIA3 | 1569290_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6214)
GRIA3 | 1569290_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRIA3 | 1569290_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0357
p=0.7084
p=0.1905
p=0.878
p=0.6248
p=0.6153
p=0.8252
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2